

DARES Propulsion & Motor Control

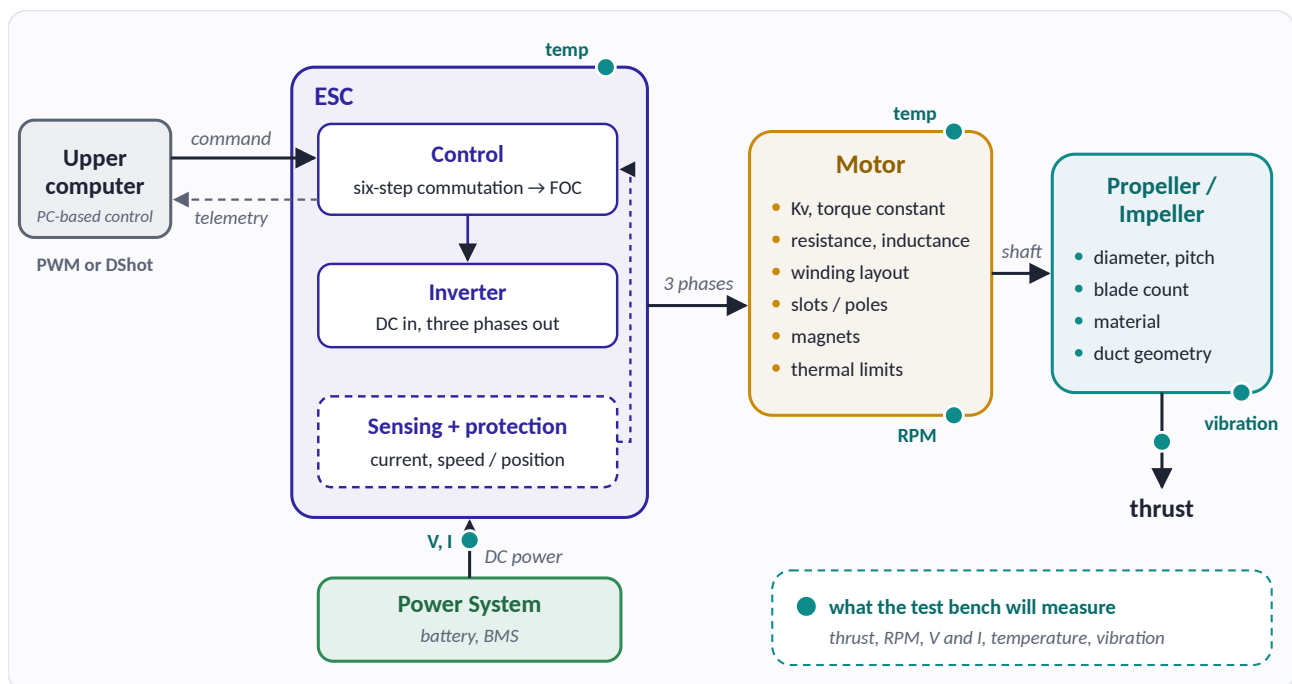
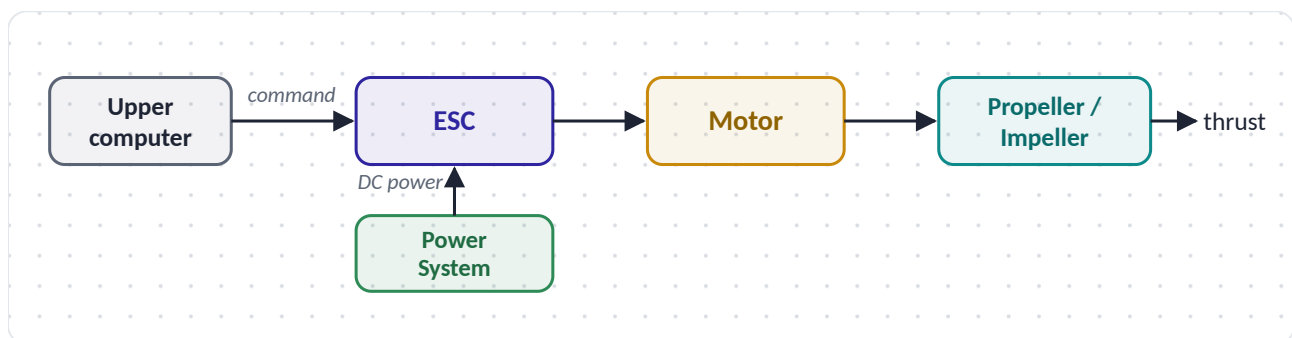
Semester Direction and Mid-Semester Break Plan

Meeting: Saturday 26 September 2026

1 What I think we should build

My proposal is that we work towards designing and testing our own propulsion system: the **motor, ESC and propeller/impeller system**. The aim is not to immediately build a perfect custom motor, but to build the tools and understanding that let us keep changing parts, testing them, and learning what actually improves performance.

As I am also learning it properly as we go I want this to be a team where everyone gets to design, simulate, build and test real things. My role is mainly to keep the work coordinated, make sure the interfaces with other teams are clear, organise equipment and approvals, and make sure our testing is safe.



The same chain, opened up. Teal dots mark what the test bench will measure.

2 How I think we should start

Because we are away over the mid-semester break, the first step should be **simulation and research rather than hardware**. We can start with simple models and gradually make them more detailed.

- Model the chain from electrical input → ESC → motor → propeller → thrust.

- Learn which motor parameters matter: K_v , torque constant, resistance, inductance, winding layout, slots/poles, magnets and thermal limits.
- Compare ESC approaches such as six-step commutation and FOC, and understand the basic inverter, sensing and protection hardware.
- Begin looking at propeller/impeller variables such as diameter, pitch, blade count, material and duct geometry.

The point is not to perfectly simulate the final drone. It is to work out what is worth testing before we spend money or start spinning hardware at high speed.

3 Main semester project: propulsion test bench

Once we are back, I think our main shared project should be a **modular propulsion test bench**. We should be able to mount a configuration, run the same test, record the results, change one thing, and run it again.

A first version should measure:

- thrust, RPM, voltage and current;
- motor and ESC temperature;
- basic vibration if practical;
- all data in a form we can save and compare.

The rig should let us experiment with different ESC designs/firmware, motor windings or geometries, propellers/impellers, materials, ducts/housings and cooling ideas. We can begin with known commercial parts to validate the rig, then progressively replace individual parts with our own designs.

Safety will be designed into the bench from the start: secure mounting, guarding/containment, remote operation, power isolation, defined RPM/current limits and staff approval before high-energy testing.

4 Team and break plan

I would like the work to stay open-ended enough that new members can join based on what they find interesting.

- **Jayden – lead:** overall motor/system direction, simulation, test-bench requirements, coordination with staff and other sub-teams, safety and approvals.
- **Jinghan – lead-old:** ESC architecture, PC-based control, sensing, protection and future custom ESC workcontrol simulation. overall motor/system direction, simulation, test-bench requirements, safety and approvals.
- **New members:** choose a real problem to own — motor design, propeller/duct design, test stand mechanics, sensing/data logging, simulation, thermal design or safety hardware:
 - reproduce/extend the propulsion model;
 - research a motor geometry or winding question;
 - model a propeller/duct idea;
 - design part of the test stand;
 - investigate sensing/data acquisition;
 - help define the safety enclosure and operating procedure.

Over the break, the goal is simply to come back with a better model of the system, a proposed test-bench design, and a few directions we want to test first.

The goal for next semester is to build the capability to design propulsion systems ourselves: simulate them, build them, measure them, change them, and understand why they got better or worse.

5 Early-stage resources

TO-DO

- **MathWorks Motor Control video series, parts 1 to 3:** short, visual introduction to how a brushless motor works, BLDC six-step control and PWM speed control (about 30 min).
<https://www.mathworks.com/videos/series/brushless-dc-motors.html>
- **APC propeller performance data:** useful baseline data for comparing model predictions with known propellers.
<https://www.apcprop.com/technical-information/performance-data/>
- **Tyto Robotics thrust-stand examples:** useful benchmark for the measurements a UAV propulsion test bench can make.
<https://www.tytorobotics.com/pages/thrust-stands>
- **Betaflight DShot documentation:** useful overview of the modern flight-controller-to-ESC command/telemetry interface.
<https://betaflight.com/docs/development/API/Dshot>

OPTIONAL ADVANCED

- **STM32 Motor Control SDK / Workbench:** useful reference implementation for PMSM/BLDC six-step and FOC, current sensing, speed/position feedback and protection.
<https://www.st.com/en/embedded-software/x-cube-mcsdk.html>
- **ODrive documentation:** practical reference for current-controlled motor calibration, resistance/inductance measurement and encoder-based control.
<https://docs.odriverobotics.com/>
- **FEMM:** free 2D finite-element software for magnetic-field modelling; suitable for an initial motor-design workflow.
<https://www.femm.info/>
- **MathWorks Simscape Electrical:** useful if our University licence gives us access to PMSM/BLDC and inverter models.
<https://www.mathworks.com/products/simscape-electrical.html>
- **MathWorks Motor Control video series, parts 4 to 6:** field-oriented control, space vector modulation and practical FOC implementation.
<https://www.mathworks.com/videos/motor-control-part-4-understanding-field-oriented-control-1587967749983.html>